Simulación de TLC (Moivre_Laplace)

- Tomar muestras de cierto tamaño
- Media de las muestras
- histograma (representa a la normalidad)

In [15]:

```
import numpy.random as npr
import numpy as np
import pandas as pd
import seaborn as sns
```

In [6]:

```
npr.seed(1)

muestras_binomiales = npr.binomial(3, 0.6, 1000)
muestras_binomiales
```

Out[6]:

```
array([2, 1, 3, 2, 3, 3, 3, 2, 2, 2, 2, 1, 3, 1, 3, 1, 2, 2, 3, 3, 1, 0,
       2, 1, 1, 1, 3, 3, 3, 1, 3, 2, 0, 2, 1, 2, 1, 1, 3, 1, 0, 1, 2, 1,
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       1, 2, 0, 2, 1, 3, 3, 1, 2, 3, 1, 2, 1, 1, 1, 2, 1, 2, 2, 1, 2, 0,
       1, 2, 3, 0, 2, 2, 2, 2, 1, 2, 3, 2, 2, 2, 1, 2, 1, 2, 3, 1, 1, 0,
       3, 3, 1, 1, 3, 1, 1, 1, 1, 3, 3, 3, 3, 2, 1, 2, 2, 1, 3, 2, 2, 0,
       2, 3, 1, 2, 1, 2, 1, 1, 2, 3, 3, 3, 3, 3, 2, 1, 2, 3, 3, 0, 2, 3,
       2, 1, 3, 2, 0, 1, 2, 2, 2, 1, 3, 3, 3, 2, 2, 2, 2, 0, 2, 2, 2, 1,
       1, 2, 3, 2, 2, 3, 1, 3, 2, 1, 3, 2, 2, 1, 2, 3, 1, 1, 1, 1, 3, 2,
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       2, 2, 1, 1, 2, 3, 3, 2, 1, 3, 2, 1, 2, 2, 2, 3, 2, 2, 3, 2, 2, 1,
       3, 1, 2, 2, 1, 3, 2, 1, 1, 2, 1, 0, 1, 1, 3, 1, 1, 2, 1, 3, 1, 1,
```

```
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0, 2, 2, 1, 2, 1, 2, 2, 2, 2, 2, 1, 3, 3, 1, 2, 2, 2, 2, 3, 1, 0,
1, 2, 2, 2, 2, 0, 2, 1, 2, 1], dtype=int32)
```

In [12]:

```python
df= pd.DataFrame()

for i in range(1,100):
    muestras_binomiales = npr.binomial(1000, 0.6, 50)
    col = f"muestras_{i}"
    df[col] = muestras_binomiales

    df_media =  pd.DataFrame(df.mean(), columns = ["muestras de las medias"])
```

In [13]:

```python
df_media
```

Out[13]:

	muestras de las medias
muestras_1	602.52
muestras_2	600.28
muestras_3	600.82
muestras_4	601.66
muestras_5	595.38
...	...
muestras_95	601.56
muestras_96	602.84
muestras_97	604.34
muestras_98	597.54
muestras_99	606.10

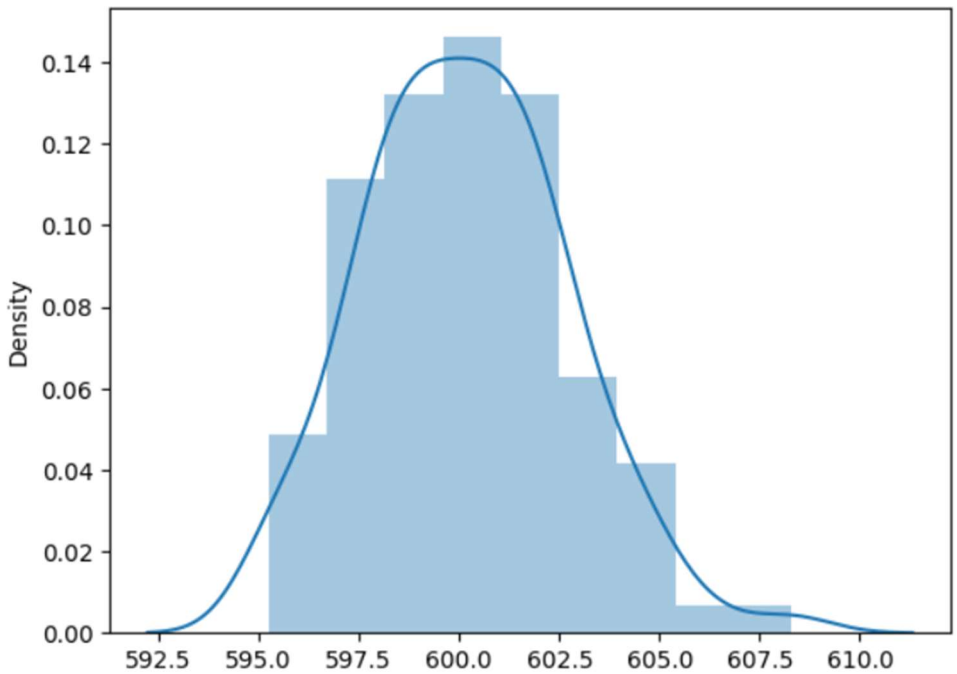
Activar Windows
Ve a Configuración para a

99 rows × 1 columns

In [16]:
```
sns.distplot(df_media)
```

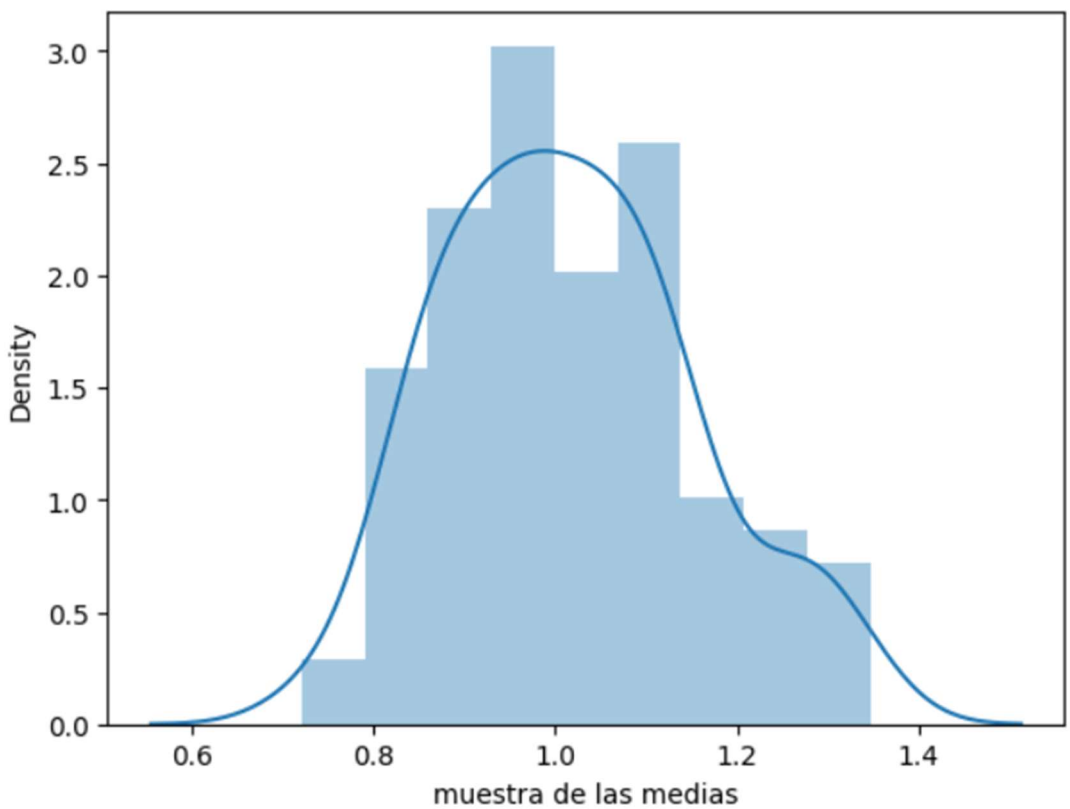
```
C:\Users\PERSONAL\AppData\Local\Temp\ipykernel_10340\2728038310.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df_media)
```

Out[16]: <Axes: ylabel='Density'>



In [22]:

```
import seaborn as sns

df = pd.DataFrame()

for i in range(1, 101):
    muestras_exponenciales = npr.exponential(1, 50)
    col = f"muestras_{i}"
    df[col] = muestras_exponenciales

df_media = pd.DataFrame(df.mean(), columns=["muestra de las medias"])

sns.distplot(df_media["muestra de las medias"])
```

```
C:\Users\PERSONAL\AppData\Local\Temp\ipykernel_10340\227469803.py:12: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df_media["muestra de las medias"])
```

Out[22]: <Axes: xlabel='muestra de las medias', ylabel='Density'>